Tele2US-2026-03-27-1

March 27, 2026

1 Task 1. Which year had the highest amount of crimes?

[ ]:
```
%%bigquery
SELECT
  EXTRACT(YEAR FROM `Date`) AS year,
  COUNT(*) AS n_crimes
FROM `alva-coding-test.chicago_crime.crime`
GROUP BY year
ORDER BY n_crimes
DESC LIMIT 1  -- 5
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:

	year	n_crimes
0	2002	486825

2 Task 2. Arrest Rate

Let's define "Arrest Rate" as the share of crimes that led to an arrest.

What year had the highest arrest rate?

What is the overall trend in number of crimes per year?

2.1 Arrest rate per year

[ ]:
```
%%bigquery
SELECT
  EXTRACT(YEAR FROM `Date`) AS year,
  COUNT(*) AS n_crimes, COUNT(CASE WHEN `arrest` = TRUE THEN 1 END) AS␣
 ↪ARRESTED_COUNT,
  COUNT(CASE WHEN `arrest` = FALSE THEN 1 END) AS NON_ARRESTED_COUNT,
  AVG(CAST(arrest AS INT64)) AS ARREST_RATE
FROM `alva-coding-test.chicago_crime.crime`
GROUP BY year
ORDER BY year;
```

```
Query is running:   0%|          |
```

```
Downloading:   0%|          |
```

[ ]:

	year	n_crimes	ARRESTED_COUNT	NON_ARRESTED_COUNT	ARREST_RATE
0	2001	485920	141942	343978	0.292110
1	2002	486825	141578	345247	0.290819
2	2003	475989	141600	334389	0.297486
3	2004	469432	144704	324728	0.308253
4	2005	453779	140928	312851	0.310565
5	2006	448191	135426	312765	0.302161
6	2007	437093	131881	305212	0.301723
7	2008	427196	110026	317170	0.257554
8	2009	392842	110840	282002	0.282149
9	2010	370540	100554	269986	0.271372
10	2011	352016	96299	255717	0.273564
11	2012	336336	90678	245658	0.269605
12	2013	307568	86567	221001	0.281456
13	2014	275846	79656	196190	0.288770
14	2015	264849	70057	194792	0.264517
15	2016	269911	53048	216863	0.196539
16	2017	269186	52666	216520	0.195649
17	2018	269035	53892	215143	0.200316
18	2019	261494	56251	205243	0.215114
19	2020	212462	34149	178313	0.160730
20	2021	209319	26535	182784	0.126768
21	2022	239489	28038	211451	0.117074
22	2023	262395	31661	230734	0.120662
23	2024	155673	20609	135064	0.132386

2.2 Year with the highest arrest rate

[ ]:

```
%%bigquery
SELECT
  EXTRACT(YEAR FROM `Date`) AS year,
  AVG(CAST(arrest AS INT64)) AS ARREST_RATE
FROM `alva-coding-test.chicago_crime.crime`
GROUP BY year
ORDER BY ARREST_RATE DESC
LIMIT 1
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:

	year	ARREST_RATE
0	2005	0.310565

2.3 Overall trend in number of crimes per year

[ ]:
```
%%bigquery df2
-- Save in a df2 dataframe
SELECT
  EXTRACT(YEAR FROM `Date`) as year,
  AVG(CAST(arrest AS INT64)) AS ARREST_RATE
FROM `alva-coding-test.chicago_crime.crime`
GROUP BY year
ORDER BY year
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:
```
display(df2)  # we've already seen it as a table, now as a dataframe
```

	year	ARREST_RATE
0	2001	0.292110
1	2002	0.290819
2	2003	0.297486
3	2004	0.308253
4	2005	0.310565
5	2006	0.302161
6	2007	0.301723
7	2008	0.257554
8	2009	0.282149
9	2010	0.271372
10	2011	0.273564
11	2012	0.269605
12	2013	0.281456
13	2014	0.288770
14	2015	0.264517
15	2016	0.196539
16	2017	0.195649
17	2018	0.200316
18	2019	0.215114
19	2020	0.160730
20	2021	0.126768
21	2022	0.117074
22	2023	0.120662
23	2024	0.132386

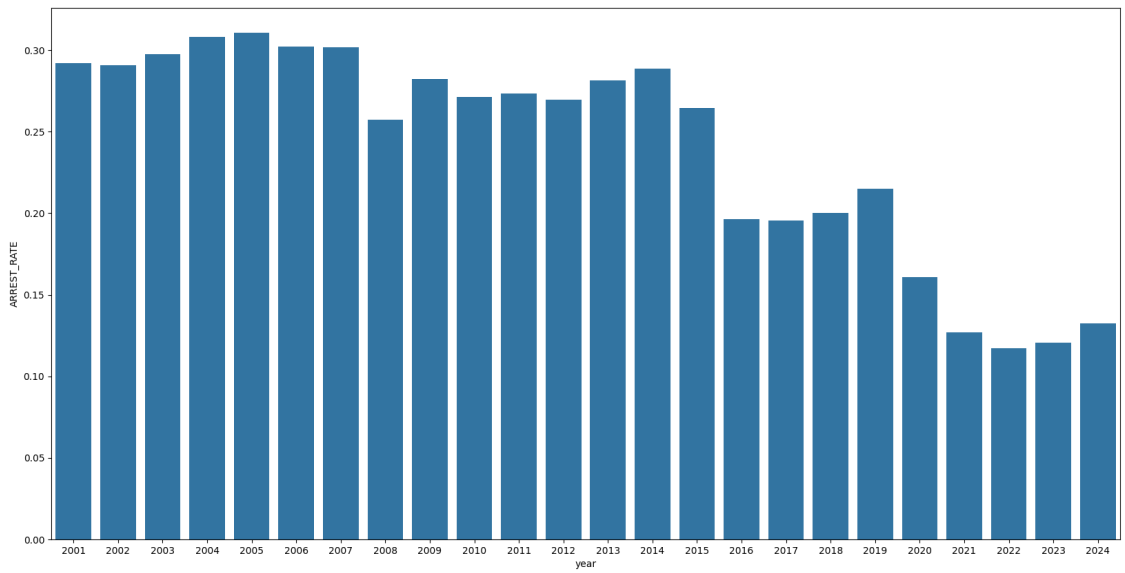
2.4 Plot the trend of the arrest rate - note the overall declining rate

[ ]:
```
# For Python graphics/diagrams
!pip install matplotlib seaborn
```

[ ]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# import sklearn  # same as scikit-learn
# import scipy.stats as stats
```

[ ]:
```
plt.figure(figsize=(20, 10))
sns.barplot(x="year", y="ARREST_RATE", data=df2)
```

[ ]: <Axes: xlabel='year', ylabel='ARREST_RATE'>



2.5 Trend in the number of crimes

[ ]:
```
%%bigquery df2a
SELECT
  EXTRACT(YEAR FROM `Date`) as year,
  COUNT(*) AS n_crimes
FROM `alva-coding-test.chicago_crime.crime`
GROUP BY year
ORDER BY year
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:
```
display(df2a)
```

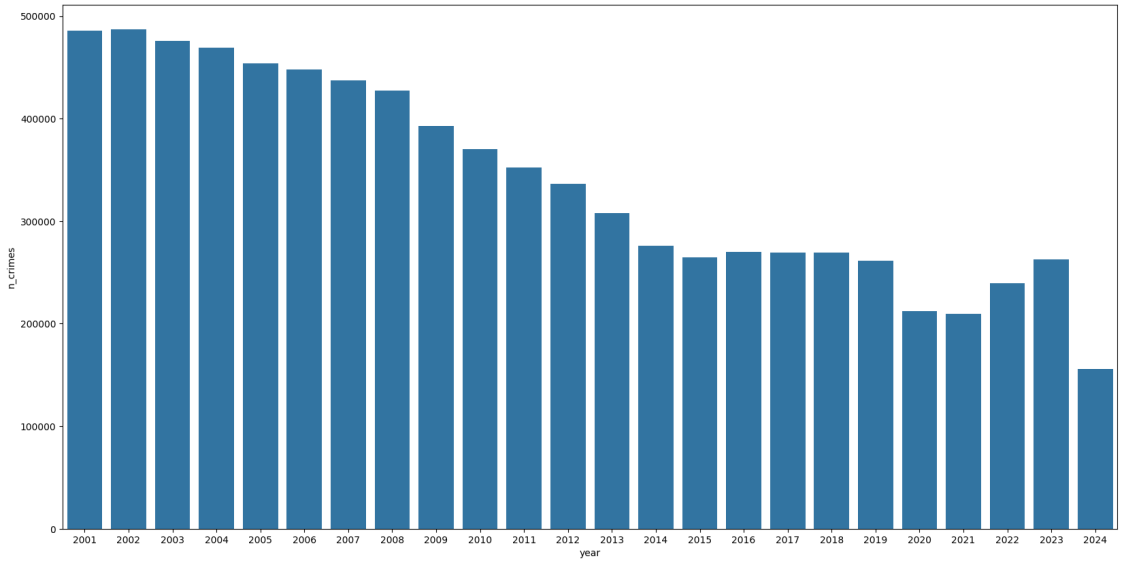
	year	n_crimes
0	2001	485920
1	2002	486825

2	2003	475989
3	2004	469432
4	2005	453779
5	2006	448191
6	2007	437093
7	2008	427196
8	2009	392842
9	2010	370540
10	2011	352016
11	2012	336336
12	2013	307568
13	2014	275846
14	2015	264849
15	2016	269911
16	2017	269186
17	2018	269035
18	2019	261494
19	2020	212462
20	2021	209319
21	2022	239489
22	2023	262395
23	2024	155673

2.6 Number of crimes per year declines

[ ]:
```
plt.figure(figsize=(20, 10))
sns.barplot(x="year", y="n_crimes", data=df2a)
```

[ ]: <Axes: xlabel='year', ylabel='n_crimes'>



3 Task 3. What were the five most common crimes in 2020?

Which of those crimes had the highest and lowest arrest rate?

[ ]:
```
%%bigquery
SELECT
  `primary_type`,
  COUNT(*) AS CRIME_COUNT,
  AVG(CAST(arrest AS INT64)) AS ARREST_RATE
FROM `alva-coding-test.chicago_crime.crime`
WHERE EXTRACT(YEAR FROM `Date`) = 2020
GROUP BY `primary_type`
ORDER BY CRIME_COUNT
DESC LIMIT 5;
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:

	primary_type	CRIME_COUNT	ARREST_RATE
0	BATTERY	41514	0.166691
1	THEFT	41337	0.067833
2	CRIMINAL DAMAGE	24877	0.047393
3	DECEPTIVE PRACTICE	18487	0.020825
4	ASSAULT	18257	0.127677

3.1 Simultaneous top 5 and Min/Max arrest rates among them

[ ]:
```
%%bigquery
WITH top_5_crimes AS (
  SELECT
    `primary_type`,
    COUNT(*) AS CRIME_COUNT,
    AVG(CAST(arrest AS INT64)) AS ARREST_RATE
  FROM `bigquery-public-data.chicago_crime.crime`
  WHERE EXTRACT(YEAR FROM `Date`) = 2020
  GROUP BY `primary_type`
  ORDER BY CRIME_COUNT DESC
  LIMIT 5
)

SELECT *, 'TOP_5' AS Legend
FROM top_5_crimes

UNION ALL

SELECT * FROM (
  (SELECT *, 'MIN_ARREST_RATE' AS Legend FROM top_5_crimes ORDER BY ARREST_RATE␣
↪ASC LIMIT 1)
```

```
  UNION ALL
  (SELECT *, 'MAX_ARREST_RATE' AS Legend FROM top_5_crimes ORDER BY ARREST_RATE␣
 ↪DESC LIMIT 1)
)
ORDER BY legend DESC; -- put TOP_5 first first
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:

	primary_type	CRIME_COUNT	ARREST_RATE	Legend
0	BATTERY	41519	0.166863	TOP_5
1	THEFT	41355	0.067803	TOP_5
2	CRIMINAL DAMAGE	24882	0.047464	TOP_5
3	DECEPTIVE PRACTICE	18634	0.020876	TOP_5
4	ASSAULT	18260	0.127711	TOP_5
5	DECEPTIVE PRACTICE	18634	0.020876	MIN_ARREST_RATE
6	BATTERY	41519	0.166863	MAX_ARREST_RATE

4 Task 4. How has the arrest rate looked like over time?

Plot the trend of the arrest rate.

Between which years can you see the biggest change in "Arrest Rate"?

Can you point at specific reasons to why the Arrest Rate dropped between those years? Comment on your conclusions.

Since the data set is constantly updating, disregard the year 2021 and later for the analysis in this question.

[ ]:
```
%%bigquery df4
SELECT
  EXTRACT(YEAR FROM `Date`) as year,
  AVG(CAST(arrest AS INT64)) AS ARREST_RATE
FROM `alva-coding-test.chicago_crime.crime`
WHERE YEAR <= 2020  -- as required
GROUP BY year
ORDER BY year
```

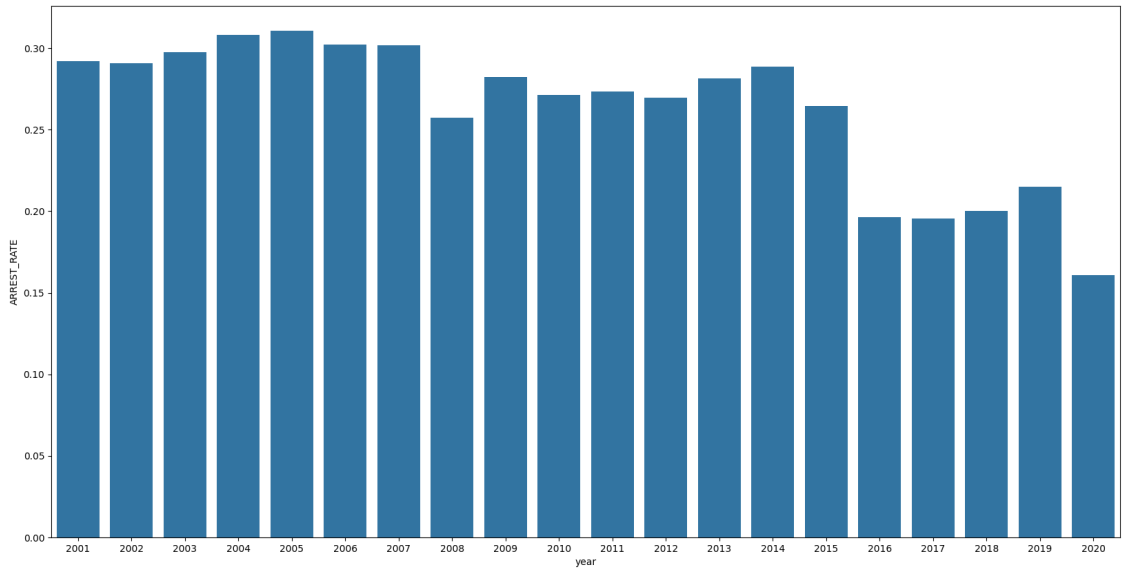
```
Query is running:   0%|          |

Downloading:   0%|          |
```

4.1 Plot the arrest rate

[ ]:
```
plt.figure(figsize=(20, 10))
sns.barplot(x="year", y="ARREST_RATE", data=df4)
```

[ ]:
```
<Axes: xlabel='year', ylabel='ARREST_RATE'>
```



4.2 Between which years can you see the biggest change in "Arrest Rate"?

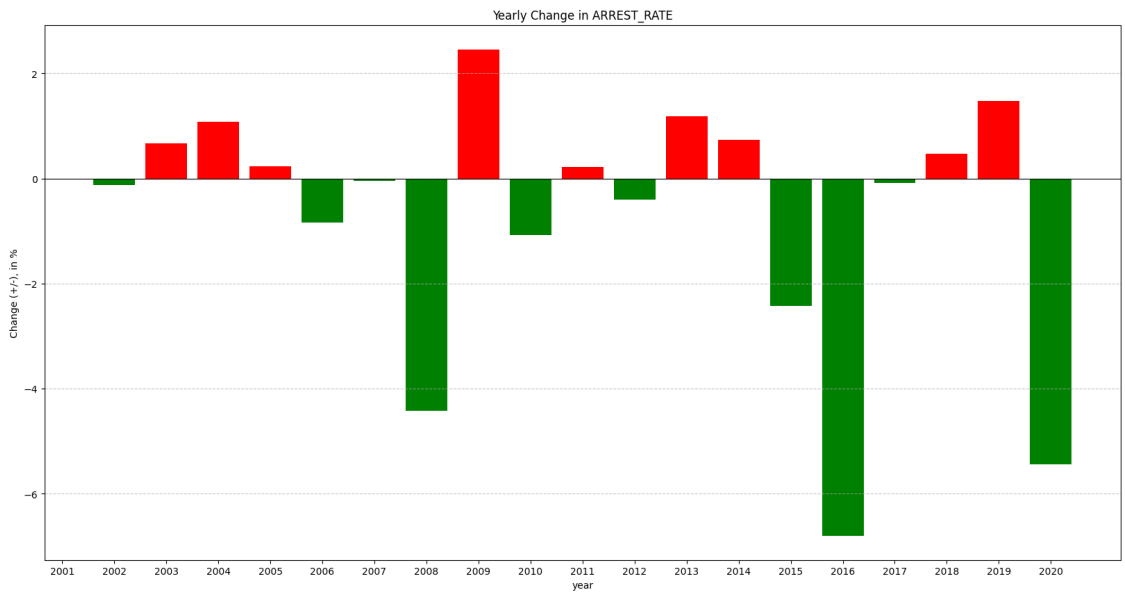
[ ]:
```
# Calculate change (this year's value minus last year's), the first/0-th is NaN␣
↪as it has no predecessor
df4['CHANGE'] = df4['ARREST_RATE'].diff()
df4
```

[ ]:

	year	ARREST_RATE	CHANGE
0	2001	0.292110	NaN
1	2002	0.290819	-0.001291
2	2003	0.297486	0.006667
3	2004	0.308253	0.010768
4	2005	0.310565	0.002312
5	2006	0.302161	-0.008404
6	2007	0.301723	-0.000438
7	2008	0.257554	-0.044169
8	2009	0.282149	0.024595
9	2010	0.271372	-0.010778
10	2011	0.273564	0.002193
11	2012	0.269605	-0.003959
12	2013	0.281456	0.011851
13	2014	0.288770	0.007313
14	2015	0.264517	-0.024253
15	2016	0.196539	-0.067978
16	2017	0.195649	-0.000890
17	2018	0.200316	0.004667
18	2019	0.215114	0.014798
19	2020	0.160730	-0.054384

4.3 Plot the year-by-year changes

[ ]:
```
plt.figure(figsize=(20, 10))
colors = ['red' if x >= 0 else 'green' for x in df4['CHANGE']]
plt.bar(df4['year'].astype(str), df4['CHANGE']*100, color=colors)  # in percent
plt.axhline(0, color='black', linewidth=0.8)  # add baseline at 0
plt.title('Yearly Change in ARREST_RATE')
plt.xlabel('year')
plt.ylabel('Change (+/-), in %')
plt.grid(axis='y', linestyle='--', alpha=0.7)
```



4.4 Biggest changes (by percent points), confirms the diagram above

[ ]:
```
row_min = df4.nsmallest(1, 'CHANGE')
row_max = df4.nlargest(1, 'CHANGE')
min_max_df = pd.concat([row_min, row_max])
min_max_df['Legend'] = ['MAX DECREASE', 'MAX INCREASE']
display(min_max_df)
```

	year	ARREST_RATE	CHANGE	Legend
15	2016	0.196539	-0.067978	MAX DECREASE
8	2009	0.282149	0.024595	MAX INCREASE

4.4.1 From -6.79% to +2.45%

4.5 Explanations:

1. The heightened arrest and incarceration numbers in Chicago around 2009 were driven by intensive "poverty management" strategies, such as the targeting of specific "hot spot" areas

on the city's south and west sides. These areas saw increased number of people committed to prison sentences. Furthermore, this period was part of a national trend, as state prison populations peaked in 2009, driven by policies like "zero tolerance" policing and mandatory minimum sentencing.

2. The significant decrease in Chicago's arrest rate in 2016, occurring despite a sharp rise in homicides and violence, was largely driven by a reduction in proactive policing following intense scrutiny of the police department. This included the aftermath of the Laquan McDonald shooting video release (late 2015) and investigations into police misconduct, leading to a "depolicing" effect where officers were less active in making stops and arrests.

4.6 Importand remark: what is more significant - increase by 2% from 10%, or increase by 1% from 1%? (the first is 20% relative vs second is 100% relative)

[ ]:
```
# Compute relative changes:
df4['REL_CHANGE'] = df4['ARREST_RATE'].pct_change()
df4
```

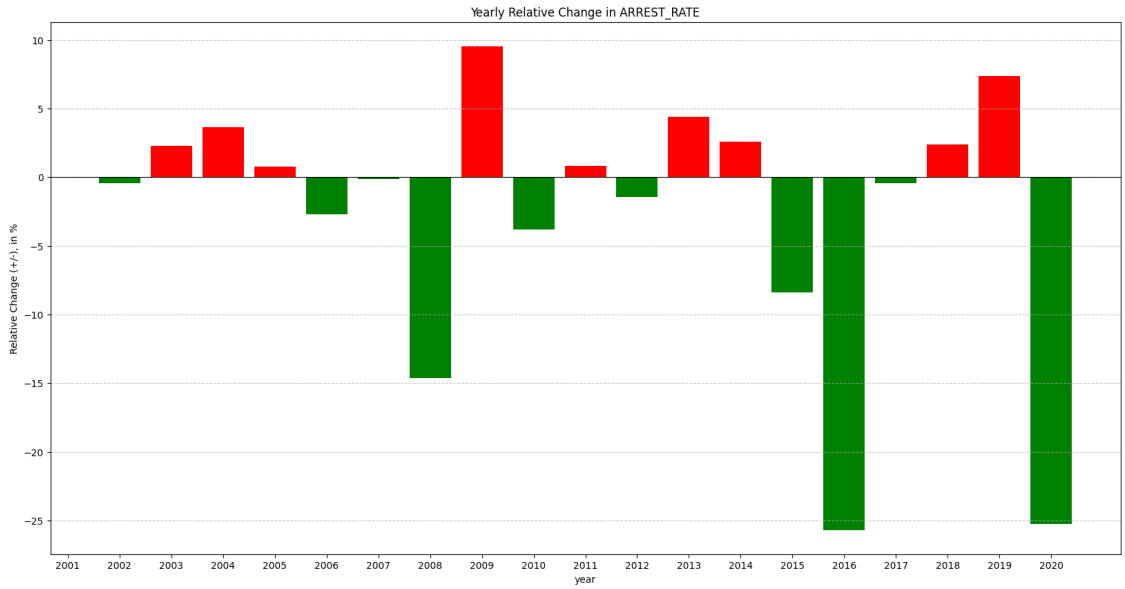
[ ]:

	year	ARREST_RATE	CHANGE	REL_CHANGE
0	2001	0.292110	NaN	NaN
1	2002	0.290819	-0.001291	-0.004419
2	2003	0.297486	0.006667	0.022924
3	2004	0.308253	0.010768	0.036195
4	2005	0.310565	0.002312	0.007500
5	2006	0.302161	-0.008404	-0.027060
6	2007	0.301723	-0.000438	-0.001451
7	2008	0.257554	-0.044169	-0.146389
8	2009	0.282149	0.024595	0.095495
9	2010	0.271372	-0.010778	-0.038198
10	2011	0.273564	0.002193	0.008080
11	2012	0.269605	-0.003959	-0.014471
12	2013	0.281456	0.011851	0.043957
13	2014	0.288770	0.007313	0.025984
14	2015	0.264517	-0.024253	-0.083988
15	2016	0.196539	-0.067978	-0.256989
16	2017	0.195649	-0.000890	-0.004527
17	2018	0.200316	0.004667	0.023853
18	2019	0.215114	0.014798	0.073873
19	2020	0.160730	-0.054384	-0.252815

4.7 We see the absolute and relative arrest rate changes differ substantially

[ ]:
```
plt.figure(figsize=(20, 10))
colors = ['red' if x >= 0 else 'green' for x in df4['REL_CHANGE']]
plt.bar(df4['year'].astype(str), df4['REL_CHANGE']*100, color=colors)  # in␣
↪percent
plt.axhline(0, color='black', linewidth=0.8)  # add baseline at 0
```

```
plt.title('Yearly Relative Change in ARREST_RATE')
plt.xlabel('year')
plt.ylabel('Relative Change (+/-), in %')
plt.grid(axis='y', linestyle='--', alpha=0.7)
```



[ ]:
```
row_min_r = df4.nsmallest(1, 'REL_CHANGE')
row_max_r = df4.nlargest(1, 'REL_CHANGE')
min_max_df_r = pd.concat([row_min_r, row_max_r])
min_max_df_r['Legend'] = ['MAX RELATIVE DECREASE', 'MAX RELATIVE INCREASE']
display(min_max_df_r)
```

	year	ARREST_RATE	CHANGE	REL_CHANGE	Legend
15	2016	0.196539	-0.067978	-0.256989	MAX RELATIVE DECREASE
8	2009	0.282149	0.024595	0.095495	MAX RELATIVE INCREASE

4.7.1 From -26% to 9.5% (i.e., the relative range is BIGGER!)

5 Task 5. What was the arrest rate for thefts during 2017 and 2018?

How much did the arrest rate decrease between those two years?

Perform a t-test to determine if the decrease in arrest rate is significant at

a 95% confidence level.

Comment on your conclusions.

5.1 Decrease in theft arrest rate from 2017 to 2018:

[ ]:
```
%%bigquery df5
WITH THEFT_2017_2018 AS (
  SELECT
    EXTRACT(YEAR FROM `Date`) AS year,
    AVG(CASE WHEN `arrest` = TRUE THEN 1 ELSE 0 END) AS ARREST_RATE_THEFT,
    COUNT(*) AS THEFT_COUNT,
    COUNT(CASE WHEN `arrest` = TRUE THEN 1 END) AS ARRESTED_THEFT_COUNT,
  FROM `alva-coding-test.chicago_crime.crime`
  WHERE EXTRACT(YEAR FROM `Date`) BETWEEN 2017 AND 2018 AND `primary_type` =␣
 ↪'THEFT'
  GROUP BY year
  ORDER BY year
)
SELECT
  year,
  THEFT_COUNT,
  ARRESTED_THEFT_COUNT,
  ARREST_RATE_THEFT,
  ARREST_RATE_THEFT - COALESCE(LAG(ARREST_RATE_THEFT) OVER (ORDER BY year),␣
 ↪ARREST_RATE_THEFT) AS diff_arrest_rate_theft,
  -- ARRESTED_THEFT_COUNT / THEFT_COUNT AS fraction
FROM  THEFT_2017_2018
ORDER BY year
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:
```
display(df5)
```

```
   year  THEFT_COUNT  ARRESTED_THEFT_COUNT  ARREST_RATE_THEFT  \
0  2017        64386                  6474           0.100550
1  2018        65289                  6339           0.097091

   diff_arrest_rate_theft
0                0.000000
1               -0.003458
```

5.1.1 Performing a Statistical Test on Two Rates (Proportions)

To compare two **rates** (proportions) instead of means, one typically uses a **Two-Sample Z-Test for Proportions** rather than a standard T-test.

1. Why we don't need Mean or Standard Deviation? For a rate (like an arrest rate of 0.15), the "mean" is simply the rate itself (p), and the variance is automatically derived from the rate and the sample size using the formula: Variance $= p \times (1 - p)$.

Important: To perform this test, we **must** know the sample size (n) for each group, i.e., total number of crimes in group X (for 2017) and group Y (for 2018).

[ ]:
```
import numpy as np
from statsmodels.stats.proportion import proportions_ztest

# As given by BigQuery table (above)
# 2017        64386        6474        0.100550
# 2018        65289        6339        0.097091
# counts = np.array([6474, 6339])  # number of thefts that led to arrests␣
↪(ARRESTED_RATE_THEFT), in 2017, 2018
# nobs = np.array([64386, 65289])  # total number of thefts, in 2017, 2088
# We saved the previous query as a dataframe and can extract the numbers:

counts = np.array([int(df5['ARRESTED_THEFT_COUNT'].iloc[0]),␣
↪int(df5['ARRESTED_THEFT_COUNT'].iloc[1])])  # number of thefts that led to␣
↪arrests (ARRESTED_RATE_THEFT), in 2017, 2018
nobs = np.array([int(df5['THEFT_COUNT'].iloc[0]), int(df5['THEFT_COUNT'].
↪iloc[1])])  # total number of thefts, in 2017, 2018

# Perform the Z-test
z_stat, p_value = proportions_ztest(counts, nobs)

print(f"Z-statistic: {z_stat:.4f}")
print(f"P-value: {p_value:.4f}")

# Check for significance at 95% confidence (alpha = 0.05)
significance = 0.95
alpha = 1 - significance
print(f"Significance: {significance:.4f}, alpha: {alpha:.4f}")

if p_value < alpha:
    print("CONCLUSION: Significant difference: The rates are statistically␣
↪different. We REJECT the null hypothesis (that theft arrest rates do not␣
↪differ in 2017 and 2018)!!!")
else:
    print("CONCLUSION: No significant difference: The difference could be due␣
↪to chance. We FAIL to REJECT the null hypothesis")
# Z-statistic: 2.0867
# P-value: 0.0369
# Significance: 0.9500, alpha: 0.0500
# CONCLUSION: Significant difference: The rates are statistically different. We␣
↪REJECT the null hypothesis (that theft arrest rates do not differ in 2017␣
↪and 2018)!!!
```

```
Z-statistic: 2.0867
P-value: 0.0369
Significance: 0.9500, alpha: 0.0500
```

```
CONCLUSION: Significant difference: The rates are statistically different. We
REJECT the null hypothesis (that theft arrest rates do not differ in 2017 and
2018)!!!
```

5.2 Conclusion. Significant difference: The rates are statistically different. We REJECT the null hypothesis (that theft arrest rates do not differ in 2017 and 2018)!!!

6 Task 6. Calculate the average number of crimes committed per day for each month.

Which month has the highest respectively lowest number of crimes per day on average?

Since the data set is constantly updating, disregard the year 2021 and later for the analysis in this question.

[ ]:
```
%%bigquery
WITH YEAR_MONTH_CRIME_COUNTS AS ( -- truncate to dates to first of each month␣
↪and count crimes per each
  SELECT
    DATE_TRUNC(`Date`, MONTH) AS year_month,  -- round '2023-05-15' down to␣
↪'2023-05-01'
    COUNT(*) AS crime_count
  FROM `alva-coding-test.chicago_crime.crime`
  WHERE EXTRACT(YEAR FROM `Date`) <= 2020
  GROUP BY year_month
  ORDER BY year_month
),
WITH_ADDED_DAYS_COUNTS AS ( -- add number of days for each month (leap years␣
↪accounted for)
  SELECT *, EXTRACT(DAY FROM LAST_DAY(DATE(`year_month`), MONTH)) AS␣
↪days_in_this_month,
  FROM YEAR_MONTH_CRIME_COUNTS
),
WITH_ADDED_CRIMES_PER_DAY AS (  -- add average per day for each month
  SELECT *, crime_count / days_in_this_month AS daily_crimes
  FROM WITH_ADDED_DAYS_COUNTS
)
SELECT
  FORMAT_TIMESTAMP('%Y-%m', DATE_TRUNC(year_month, MONTH)) AS year_month, --␣
↪format YYYY-MM and add all we need
  crime_count,
  daily_crimes,
  Legend
FROM
  ((SELECT *, 'Min daily crimes' AS Legend FROM WITH_ADDED_CRIMES_PER_DAY ORDER␣
↪BY daily_crimes ASC LIMIT 1)
```

```
    UNION ALL
   (SELECT *, 'Max daily crimes' AS Legend FROM WITH_ADDED_CRIMES_PER_DAY ORDER␣
↪BY daily_crimes DESC LIMIT 1)
  )
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:

	year_month	crime_count	daily_crimes	Legend
0	2002-07	46014	1484.322581	Max daily crimes
1	2020-04	12936	431.200000	Min daily crimes

6.1 April 2020 has minimal daily crimes, July 2002 - maximum

7 Task 7. Select the five most common crimes and write and SQL query that answer the following for each crime (i.e. each primary_type).

How many crimes are committed per day on average for the month with the lowest average? Which month is this?

How many crimes are committed per day on average for the month with the highest average? Which month is it?

Since the data set is constantly updating, disregard the year 2021 and later for the analysis in this question.

7.1 Top 5 most frequent crimes

[ ]:

```
%%bigquery
SELECT `primary_type`, COUNT(*) AS n_crimes
FROM `alva-coding-test.chicago_crime.crime`
WHERE EXTRACT(YEAR FROM `Date`) <= 2020
GROUP BY `primary_type`
ORDER BY n_crimes DESC
LIMIT 5
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:

	primary_type	n_crimes
0	THEFT	1531412
1	BATTERY	1329532
2	CRIMINAL DAMAGE	826584
3	NARCOTICS	736115
4	ASSAULT	459974

7.2 Create a reusable table function for computing MIN/MAX daily averages (per month, through all months) for different kinds of primes and final year (parameters)

7.2.1 Important:

1. I am NOT allowed to create TABLE FUNCTIONs (see next cell) in `alva-coding-test.chicago_crime.crime`
2. I need the table `bigquery-public-data.chicago_crime.crime` which only exists in the US (too big to copy to my project)

[ ]:
```
%%bigquery
CREATE OR REPLACE TABLE FUNCTION `tele2-crime.crime_analysis_us.
↪min_max_monthly_crimes_averages`(crime_type STRING, end_year INT) AS (
 WITH YEAR_MONTH_CRIME_COUNTS AS (
   SELECT
     DATE_TRUNC(DATE(`Date`), MONTH) AS year_month,
     COUNT(*) AS crime_count
   -- FROM `alva-coding-test.chicago_crime.crime`  -- !!! User does not have␣
↪permission to query table alva-coding-test:chicago_crime.crime (EU)
   FROM `bigquery-public-data.chicago_crime.crime`  -- US
   WHERE EXTRACT(YEAR FROM `Date`) <= COALESCE(end_year, 2020)
     AND `primary_type` = crime_type
   GROUP BY year_month
 ),
 WITH_ADDED_DAYS_COUNTS AS (
   SELECT *, EXTRACT(DAY FROM LAST_DAY(DATE(`year_month`), MONTH)) AS␣
↪days_in_this_month
   FROM YEAR_MONTH_CRIME_COUNTS
 ),
 WITH_ADDED_CRIMES_PER_DAY AS (
   SELECT *, crime_count / days_in_this_month AS daily_crimes
   FROM WITH_ADDED_DAYS_COUNTS
 )
 SELECT
   crime_type AS requested_crime_type,
   FORMAT_DATE('%Y-%m', year_month) AS year_month_label,
   crime_count,
   daily_crimes,
   status
 FROM (
   (SELECT *, 'Lowest month' AS status FROM WITH_ADDED_CRIMES_PER_DAY ORDER BY␣
↪daily_crimes ASC LIMIT 1)
     UNION ALL
   (SELECT *, 'Highest month' AS status FROM WITH_ADDED_CRIMES_PER_DAY ORDER␣
↪BY daily_crimes DESC LIMIT 1)
 )
```

```
);
```

```
Query is running:   0%|          |
```

[ ]:
```
Empty DataFrame
Columns: []
Index: []
```

7.3 Now let us compute minima and maxima daily averages per month, for top 5 types of crimes

[ ]:
```
%%bigquery
SELECT * FROM `tele2-crime.crime_analysis_us.
↪min_max_monthly_crimes_averages`('THEFT', NULL)
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:
```
  requested_crime_type year_month_label  crime_count  daily_crimes  \
0                THEFT          2020-04         2508     83.600000
1                THEFT          2004-07        10071    324.870968

           status
0   Lowest month
1  Highest month
```

[ ]:
```
%%bigquery
SELECT * FROM `tele2-crime.crime_analysis_us.
↪min_max_monthly_crimes_averages`('BATTERY', NULL)
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:
```
  requested_crime_type year_month_label  crime_count  daily_crimes  \
0              BATTERY          2020-12         2729     88.032258
1              BATTERY          2002-06         9521    317.366667

           status
0   Lowest month
1  Highest month
```

[ ]:
```
%%bigquery
SELECT * FROM `tele2-crime.crime_analysis_us.
↪min_max_monthly_crimes_averages`('CRIMINAL DAMAGE', NULL)
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:
```
  requested_crime_type year_month_label  crime_count  daily_crimes  \
0      CRIMINAL DAMAGE          2015-02         1504     53.714286
1      CRIMINAL DAMAGE          2002-07         5413    174.612903

          status
0   Lowest month
1  Highest month
```

[ ]:
```
%%bigquery
SELECT * FROM `tele2-crime.crime_analysis_us.
↪min_max_monthly_crimes_averages`('NARCOTICS', NULL)
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:
```
  requested_crime_type year_month_label  crime_count  daily_crimes  \
0            NARCOTICS          2003-10         5821    187.774194
1            NARCOTICS          2020-04          179      5.966667

          status
0  Highest month
1   Lowest month
```

[ ]:
```
%%bigquery
SELECT * FROM `tele2-crime.crime_analysis_us.
↪min_max_monthly_crimes_averages`('ASSAULT', NULL)
```

```
Query is running:   0%|          |

Downloading:   0%|          |
```

[ ]:
```
  requested_crime_type year_month_label  crime_count  daily_crimes  \
0              ASSAULT          2014-01         1037     33.451613
1              ASSAULT          2002-06         3094    103.133333

          status
0   Lowest month
1  Highest month
```

[ ]:
```

```